

Semester	B.E. Semester VII – Computer Engineering
Subject	R Programming
Subject Professor In- charge	Prof. Sanjeev Dwivedi
Assisting Teachers	Prof. Sanjeev Dwivedi
Laboratory	M-411B

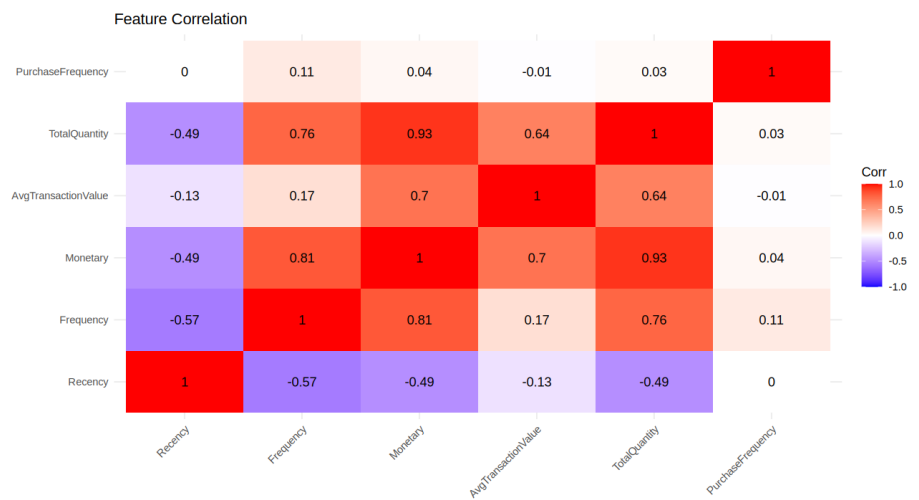
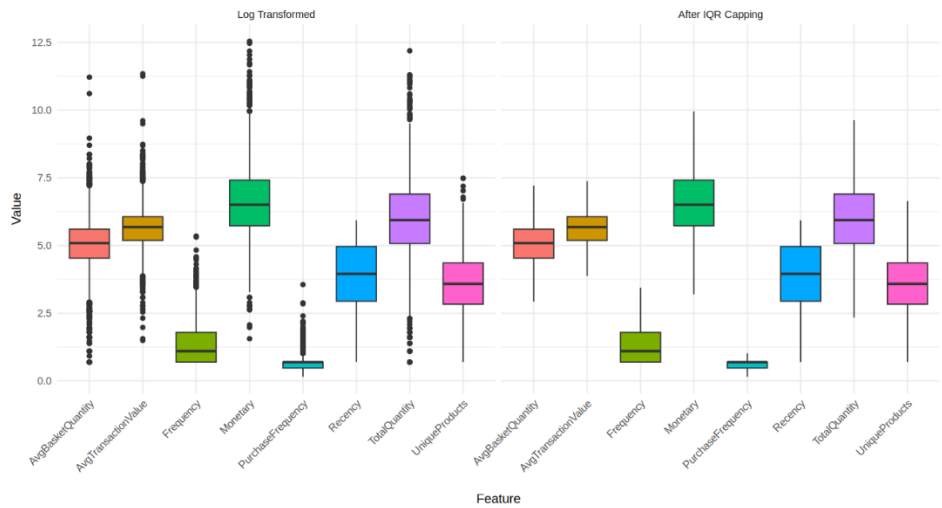
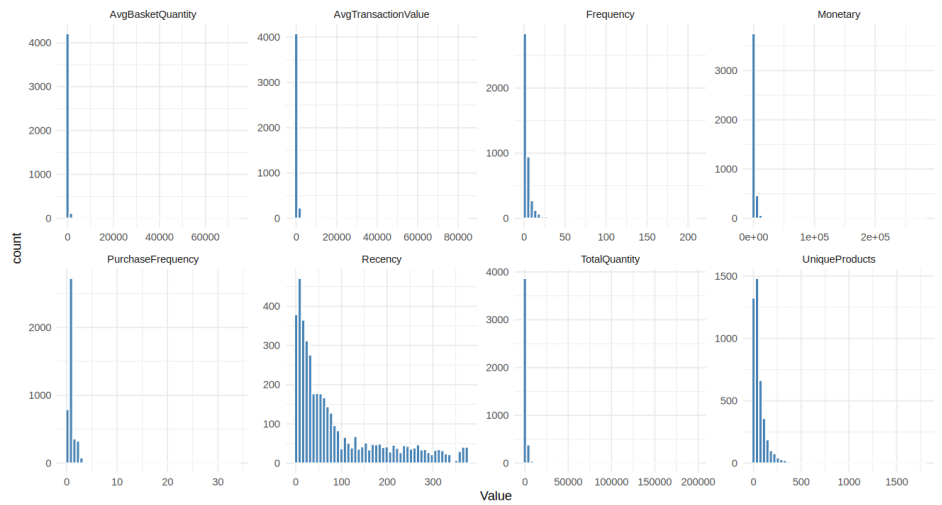
Student Name	Sneha Gadhari
Roll Number	23102B0055
Grade and Subject Teacher's Signature	

Experiment Number	07
Experiment Title	Customer segmentation and predictive analysis using machine learning

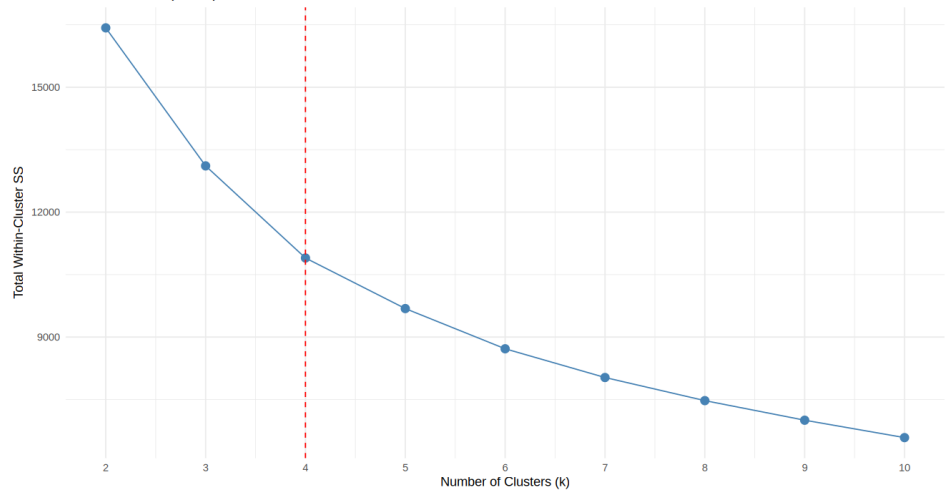
Resources / Apparatus Required	Hardware: Computer system	Software: Colab
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Objective	The objective of this experiment is to segment e-commerce customers using the UCI Online Retail dataset and to predict which customers belong to the high-value segment. Customer-level RFM and behavioural features are engineered, grouped with K-Means and Hierarchical Clustering, validated with the Silhouette Score and PCA, and then used to train Random Forest and SVM classifiers. The results are used to recommend targeted marketing strategies for each segment.
Description	The experiment was implemented in R on Google Colab using the UCI Online Retail dataset, which contains 541,909 transactions with invoice, product, quantity, date, price, customer and country attributes. The dataset is loaded directly from the UCI repository link. The workflow runs from data cleaning and feature engineering through clustering, dimensionality reduction, profiling and supervised prediction. • Data cleaning: Removed 135,080 records with missing CustomerID, 8,905 cancelled invoices, 40 rows with non-positive quantity or price, and 5,192 duplicates. This left 392,692 transactions from 4,338 customers. • Feature engineering: Built eight customer-level features: Recency, Frequency, Monetary, Average Transaction Value, Total Quantity, Purchase Frequency, Unique Products and Average Basket Quantity. • Outlier treatment and scaling: Applied a log transformation, capped outliers using the IQR rule, and standardised the clustering features. • K-Means clustering: The Elbow Method gave an optimal k of 4. The K-Means Silhouette Score was 0.2672. • Hierarchical clustering: Ward linkage with a dendrogram gave a Silhouette Score of 0.2123. The Adjusted Rand Index against K-Means was 0.5141, showing moderate agreement. • PCA: The first two components explained 73.91% of the variance (PC1 56.2%, PC2 17.71%). The 2D projection separated the clusters clearly, and an interactive 3D Plotly view was also produced. • Cluster profiles: ◦ Cluster 3, Champions: 790 customers (18.21%), average recency 20.66 days, frequency 12.92, monetary value 7,845.62. This cluster generates

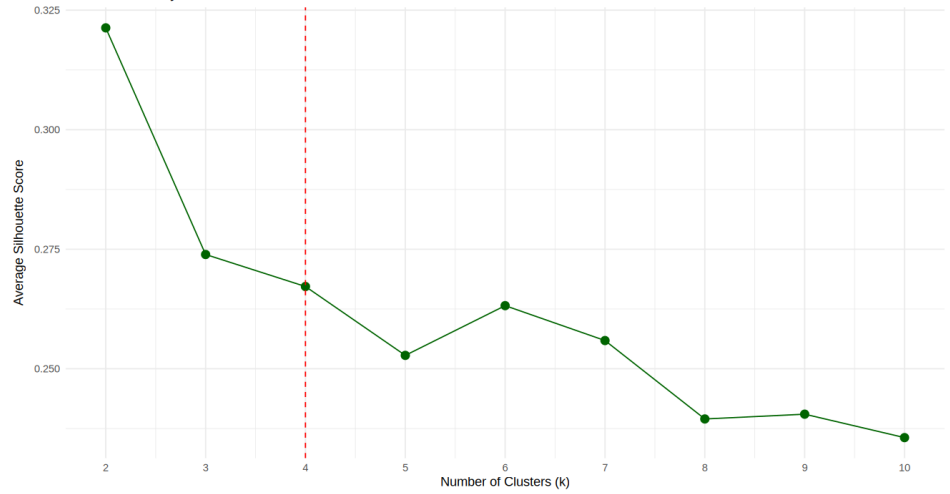
	69.74% of revenue. - Cluster 4, Recent Low-Value: 1,322 customers (30.47%), recency 51.13, frequency 3.75, monetary value 1,236.75, 18.40% of revenue. - Cluster 2, Dormant (mid-value): 1,165 customers (26.86%), recency 128.99, monetary value 732.73, 9.61% of revenue. - Cluster 1, Dormant (low-value): 1,061 customers (24.46%), recency 157.62, frequency 1.43, monetary value 189.03, 2.26% of revenue. Classification: Champions were labelled as the high-value class (18.21% of customers). Using an 80/20 stratified split, both models were tuned and compared on the test set: <table><tr><th>Model</th><th>Accuracy</th><th>Precision</th><th>Recall</th><th>F1-Score</th><th>ROC-AUC</th></tr><tr><td>Random Forest</td><td>0.9781</td><td>0.9664</td><td>0.9114</td><td>0.9381</td><td>0.9974</td></tr><tr><td>SVM (RBF)</td><td>0.9942</td><td>1.0000</td><td>0.9684</td><td>0.9839</td><td>0.9998</td></tr></table> Feature importance: Monetary value, Total Quantity and Frequency were the strongest predictors of high-value status in the Random Forest. Marketing recommendations: Champions get VIP loyalty and early-access offers. Recent low-value customers get onboarding and second-purchase incentives. Dormant segments get low-cost reactivation and clearance campaigns.	Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	Random Forest	0.9781	0.9664	0.9114	0.9381	0.9974	SVM (RBF)	0.9942	1.0000	0.9684	0.9839	0.9998																																										
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Output	CustomerID Recency Frequency Monetary TotalQuantity UniqueProducts AvgTransactionValue AvgBasketQuantity PurchaseFrequency <int> <dbl> <int> <dbl> <dbl> <int> <dbl> <dbl> <dbl> <table><tr><td>1</td><td>12346</td><td>326</td><td>1</td><td>77183.60</td><td>74215</td><td>1</td><td>77183.6000</td><td>74215.0000</td><td>1.0000000</td></tr><tr><td>2</td><td>12347</td><td>2</td><td>7</td><td>4310.00</td><td>2458</td><td>103</td><td>615.7143</td><td>351.1429</td><td>0.5752823</td></tr><tr><td>3</td><td>12348</td><td>75</td><td>4</td><td>1797.24</td><td>2341</td><td>22</td><td>449.3100</td><td>585.2500</td><td>0.4243990</td></tr><tr><td>4</td><td>12349</td><td>19</td><td>1</td><td>1757.55</td><td>631</td><td>73</td><td>1757.5500</td><td>631.0000</td><td>1.0000000</td></tr><tr><td>5</td><td>12350</td><td>310</td><td>1</td><td>334.40</td><td>197</td><td>17</td><td>334.4000</td><td>197.0000</td><td>1.0000000</td></tr><tr><td>6</td><td>12352</td><td>36</td><td>8</td><td>2506.04</td><td>536</td><td>59</td><td>313.2550</td><td>67.0000</td><td>0.9227713</td></tr></table>	1	12346	326	1	77183.60	74215	1	77183.6000	74215.0000	1.0000000	2	12347	2	7	4310.00	2458	103	615.7143	351.1429	0.5752823	3	12348	75	4	1797.24	2341	22	449.3100	585.2500	0.4243990	4	12349	19	1	1757.55	631	73	1757.5500	631.0000	1.0000000	5	12350	310	1	334.40	197	17	334.4000	197.0000	1.0000000	6	12352	36	8	2506.04	536	59	313.2550	67.0000	0.9227713
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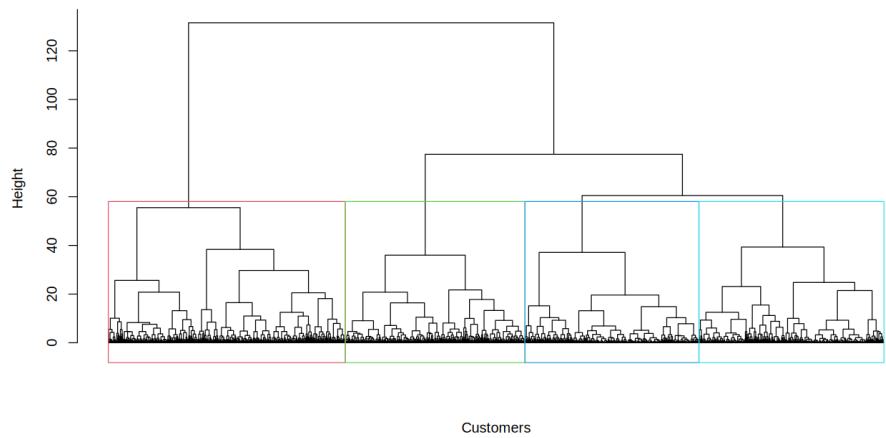
Elbow Method (k = 4)



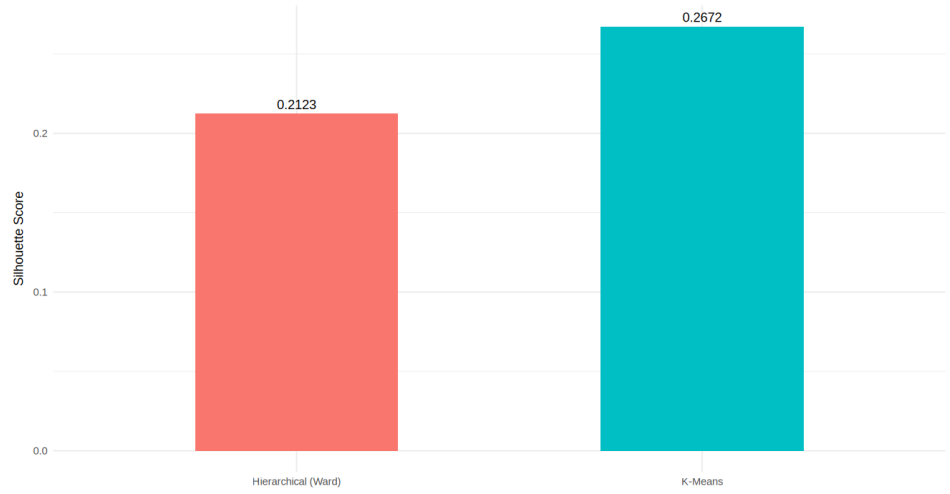
Silhouette Analysis



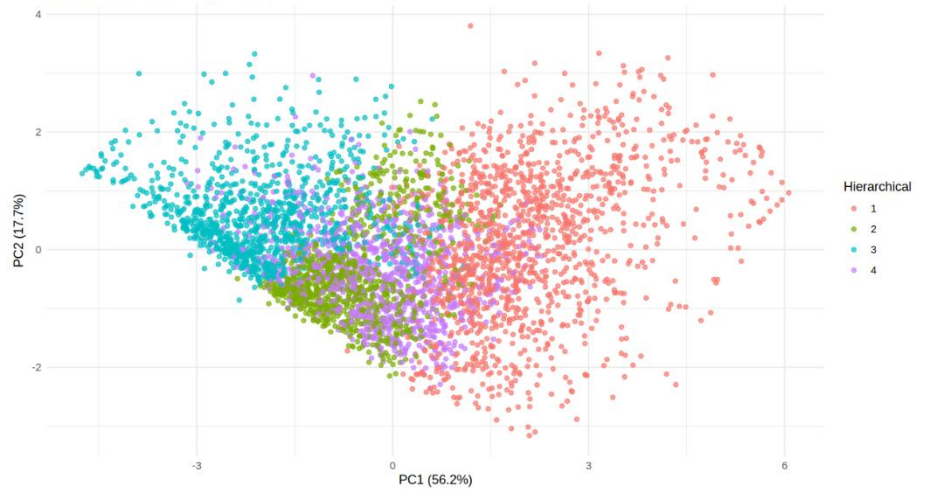
Hierarchical Clustering Dendrogram (Ward Linkage)



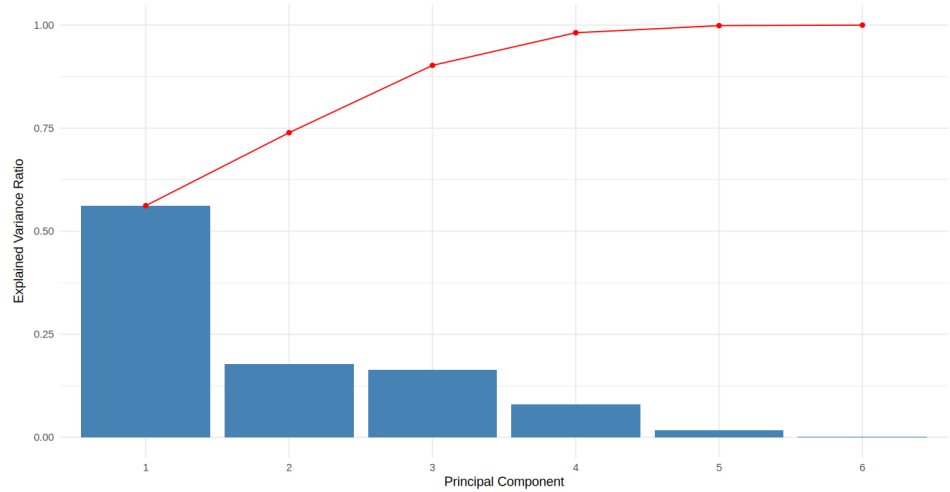
Silhouette Score Comparison

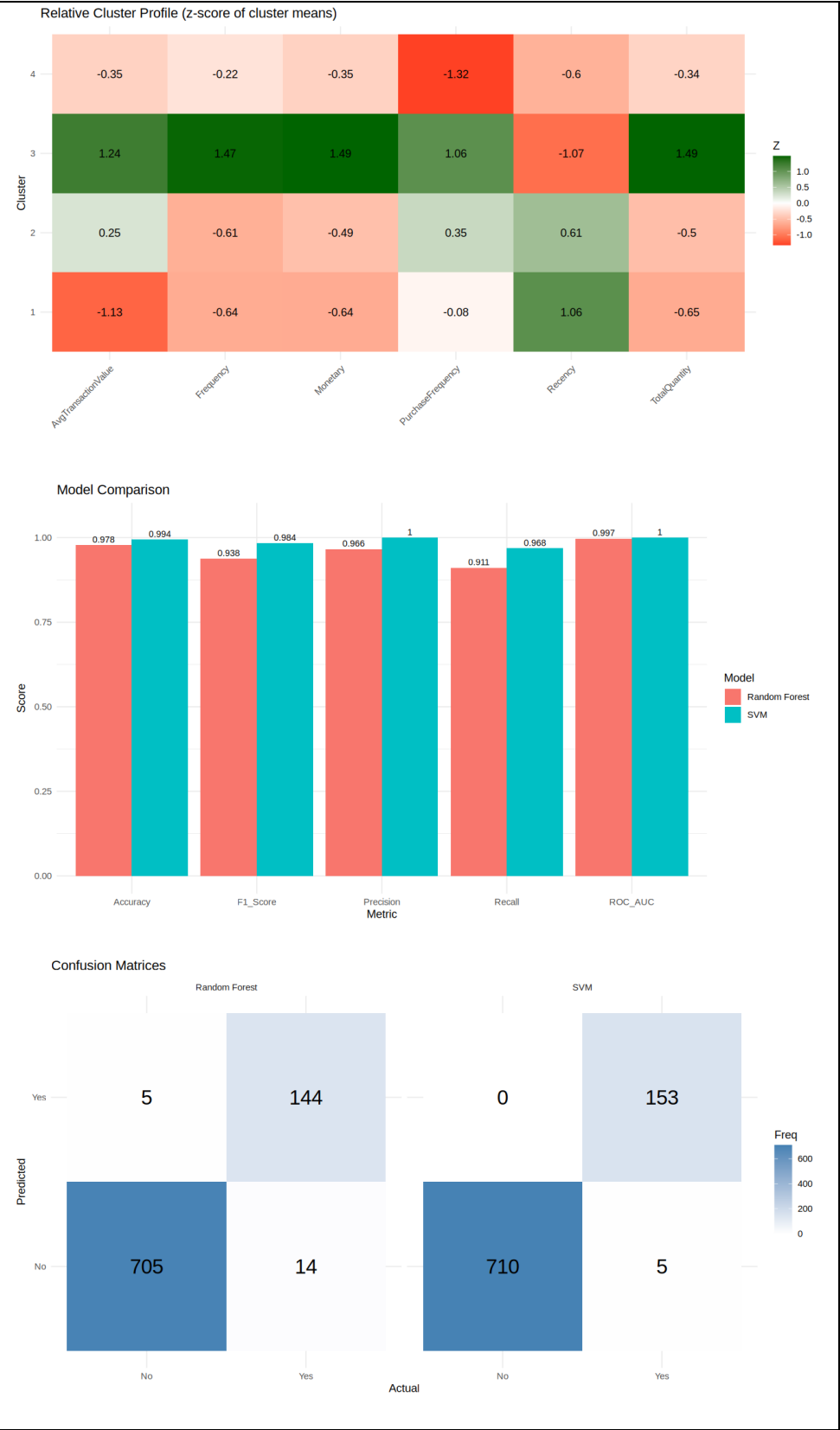


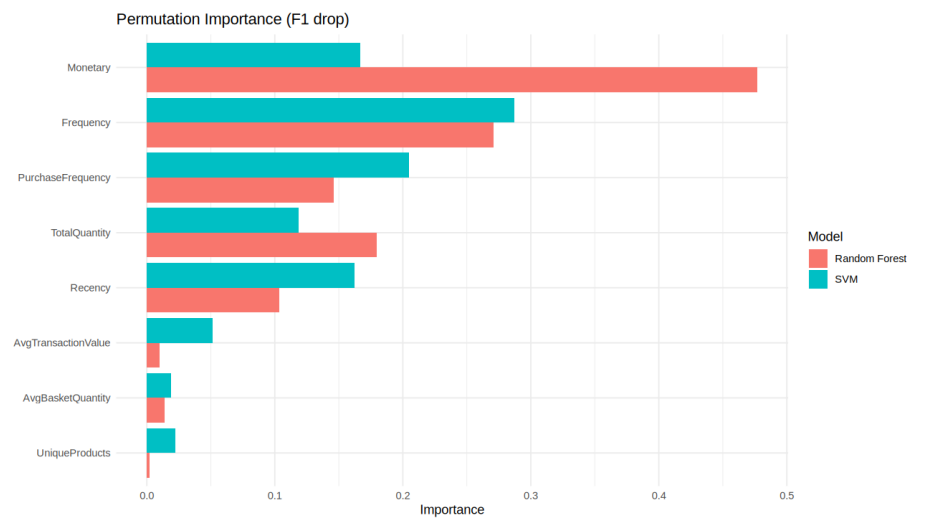
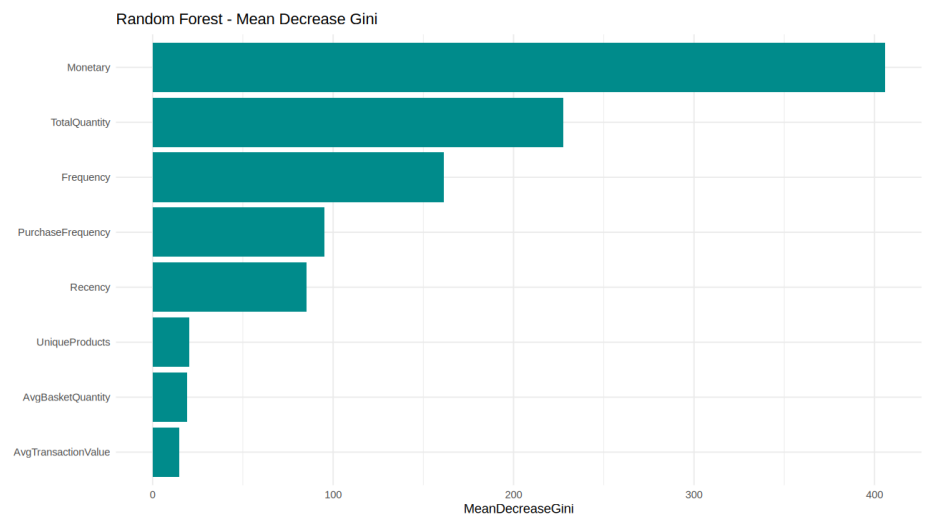
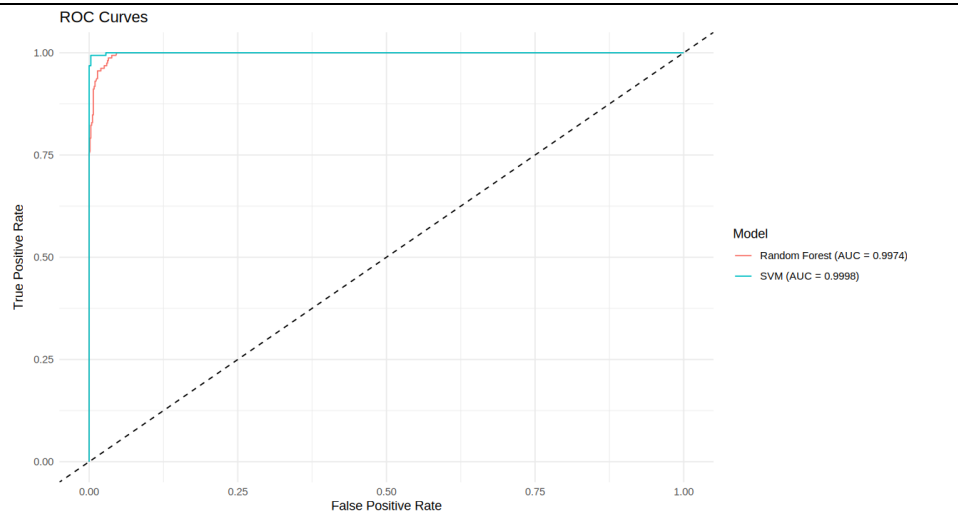
Hierarchical Segments (PCA 2D)



PCA Explained Variance







	<table><tr><th></th><th>Metric</th><th>Value</th></tr><tr><td>1</td><td>Customers Analysed</td><td>4338</td></tr><tr><td>2</td><td>Optimal k (Elbow)</td><td>4</td></tr><tr><td>3</td><td>K-Means Silhouette</td><td>0.2672</td></tr><tr><td>4</td><td>Hierarchical Silhouette</td><td>0.2123</td></tr><tr><td>5</td><td>K-Means vs Hierarchical ARI</td><td>0.5141</td></tr><tr><td>6</td><td>High-Value Cluster 3 - Champions</td><td></td></tr><tr><td>7</td><td>High-Value Customers (%)</td><td>18.21</td></tr><tr><td>8</td><td>Best Model (F1)</td><td>SVM</td></tr><tr><td>9</td><td>Best F1-Score</td><td>0.9839</td></tr><tr><td>10</td><td>Best ROC-AUC</td><td>0.9998</td></tr></table>		Metric	Value	1	Customers Analysed	4338	2	Optimal k (Elbow)	4	3	K-Means Silhouette	0.2672	4	Hierarchical Silhouette	0.2123	5	K-Means vs Hierarchical ARI	0.5141	6	High-Value Cluster 3 - Champions		7	High-Value Customers (%)	18.21	8	Best Model (F1)	SVM	9	Best F1-Score	0.9839	10	Best ROC-AUC	0.9998
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Conclusion	The experiment shows that RFM-based clustering separates customers into four meaningful segments. A small Champions group (18.21% of customers) accounts for nearly 70% of revenue. K-Means gave better-separated clusters than Hierarchical Clustering, and the SVM was the best classifier (F1 = 0.9839, ROC-AUC = 0.9998). Because the class label was derived from the clustering features, the high scores should be read as confirming cluster separability rather than as independent predictive accuracy. The segment profiles provide a practical basis for targeted marketing, retention and reactivation strategies.																																	